

Position: Rethinking Post-Hoc Search-Based Neural Approaches for Solving Large-Scale TSP

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- **Key Point:** Heatmap-guided **MCTS** underperforms **LKH-3** despite ML guidance.
- Introduce **SoftDist**: simple, unsupervised heatmap baseline.

Code
github.com/xyffffff/rethink_mcts_for_tsp

Topic
Large-Scale TSP • ML + MCTS

TitleRethinking heatmap-guided **MCTS**

Metric & Setup — Score and Evaluation

Score = (Gap_LKH-3 / Gap_MCTS) × 100% Gap_X = L_X / L* - 1, with identical CPU/time budgets.

- **Why Score:** fair relative efficiency of **MCTS** vs. **LKH-3** under same hardware and time.
- **Data:** TSP-500/1000 (1024 inst.), TSP-10000 (128 inst.), seed 1234.
- τ (grid-searched): 0.0066 (500), 0.0051 (1000), 0.0018 (10000).
- **Hardware:** Heatmaps on A100 (≈0s); **MCTS** on AMD EPYC 64C; threads match prior work.

Metric & SetupScore enables apples-to-apples vs. **LKH-3**

Problem & Method — SoftDist

- **Problem:** In large-scale TSP, ML generates edge-heatmaps to guide **MCTS**; training uses surrogate losses, testing relies on MCTS → mismatch.
- **SoftDist** method: apply softmax over negative distances with a single temperature τ to score edges; favors short edges while keeping exploration.
- **No supervision**; optimize τ by grid search using the original TSP objective with **MCTS** in the loop; same τ used at test → aligned train/test.
- **Lightweight:** negligible heatmap time; hardware-friendly.

Method**SoftDist** aligns objectives; reduces complexity

Results — Main Empirical Key Points

- **SoftDist** achieves small gaps: 1.44% (500), 2.20% (1000), 3.13% (10000) with ≈0 heatmap latency.
- **MCTS** lags **LKH-3** under equal budgets: ≪10% efficiency on 500/1000; ~1/3 on 10000 (by **Score**).
- **DIFUSCO** is competitive but far slower at heatmap generation.
- Search tuning helps all methods but **LKH-3** remains superior.



Illustrative bar chart of **Score** (TSP-500 scale for visualization; higher is better). **LKH-3** is reference (100%).

Results**SoftDist** fast and strong heatmaps; **LKH-3** dominates by **Score**

Discussion & Conclusion

- **Critique:** Surrogate-loss training misaligned with **MCTS** testing; weak theoretical grounding.
- **SoftDist:** simple, unsupervised, aligned; exposes limits of complex ML heatmaps.
- **Score** shows guided **MCTS** lags **LKH-3** under equal budgets across scales.
- **Future:** theory-grounded heatmaps and end-to-end, generalizable ML without heavy post-hoc search.